UBT514: NATURAL PRODUCTS

L	T	P	Cr
3	0	0	3.0

Course Objective: To study the drugs from a natural source, their specific method of isolation, stereochemistry, biological activity, and biogenesis/biosynthesis.

Syllabus

The general aspect of sources of medicinal plant products: Introduction to primary and secondary metabolites, types of secondary metabolites, production under stress, isolation of active constituent from plant material.

Alkaloids Definition, general properties, Classification, methods of isolation, stereochemistry, biological activity, general theory of biogenesis. Role of alkaloids in plants and their pharmaceutical importance.

Glycosides and saponins: Definition, Classification, general properties, medicinal importance, separation and isolation, structure determination, biological activity, the study of examples such as cardiac glycosides from Digitalis.

Steroids and triterpenoids: Definition, general properties, Classification, methods of isolation, biological activity, general theory of biogenesis, steroids from *Withania somnifera*, *Holarrhena* and *Solanum.*

Pigments: Occurrence, classification, introduction, and applications of carotenoids, xanthophylls, anthocyanins, flavones, flavonols. Acetate pathway and Shikimic acid pathway. Natural products of therapeutic importance from animals. Isolation, qualitative and quantitative analysis of secondary metabolite, Edible dyes, plant sweeteners, perfumery, and cosmetic agents.

Laboratory Work (if applicable)

Course Learning Objectives (CLO)
The students will be able to:

1. Apply the acquire knowledge about the secondary metabolites, natural products in various therapies.
2. Design experiments for enhanced phytochemical production.
3. Design the process of natural products isolation and purification.
4. Will have a good understanding of biosynthetic pathways of natural products.

Text Books

1. Harborne, J. B. (2013) Phytochemical Methods, Second Edition, Springer publication.
2. K. G. Ramawat and J. M. Merillon (Eds.), 2010, Biotechnology – secondary metabolites, Oxford & IBH publishing Co. Pvt. Ltd.

Reference Books

1. S. V. Bhat, B. A. Nagasampagi and M. Sivakumar 2008. Chemistry of Natural Products, First Edition, Narosa Publishing House, New Delhi
2. V. P. Agrawal and V. P. Khamboj, (Eds.) Chemistry and biology of herbal medicine:
3. G. E. Trease and W. C. Evans, 2002, Pharmacognosy and Phytochemistry, 15th Edition, W.B. Saunders Edinburgh, New York.

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments/quizzes)	30